

## Exponential regression



1

a 3

b

i Yes    ii No    iii No    iv No

2

a 4

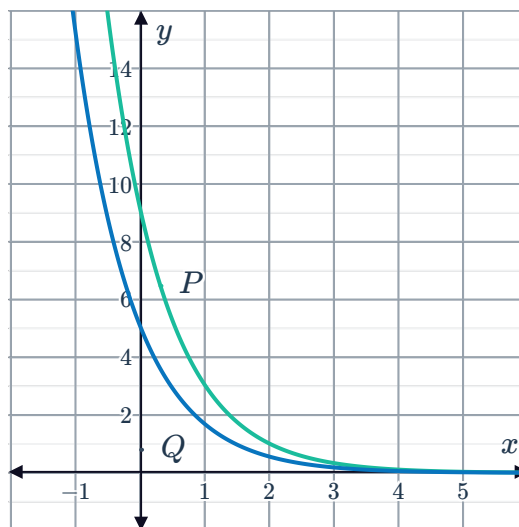
b

i No    ii Yes    iii No    iv No

3

a Decreasing

b



c Both functions approach 0.

d

i A vertical dilation by a factor of 9.

ii A vertical dilation by a factor of 5.

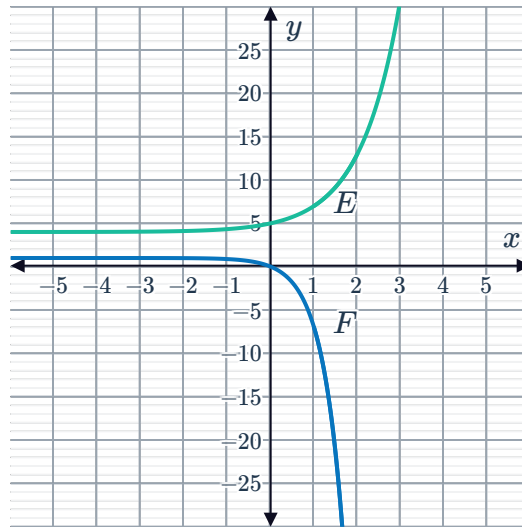
4

a Once

b Function  $P$

5

a



b Zero times

c As  $x \rightarrow \infty$ , the function  $E \rightarrow \infty$  and function  $F \rightarrow -\infty$ .

## Applications

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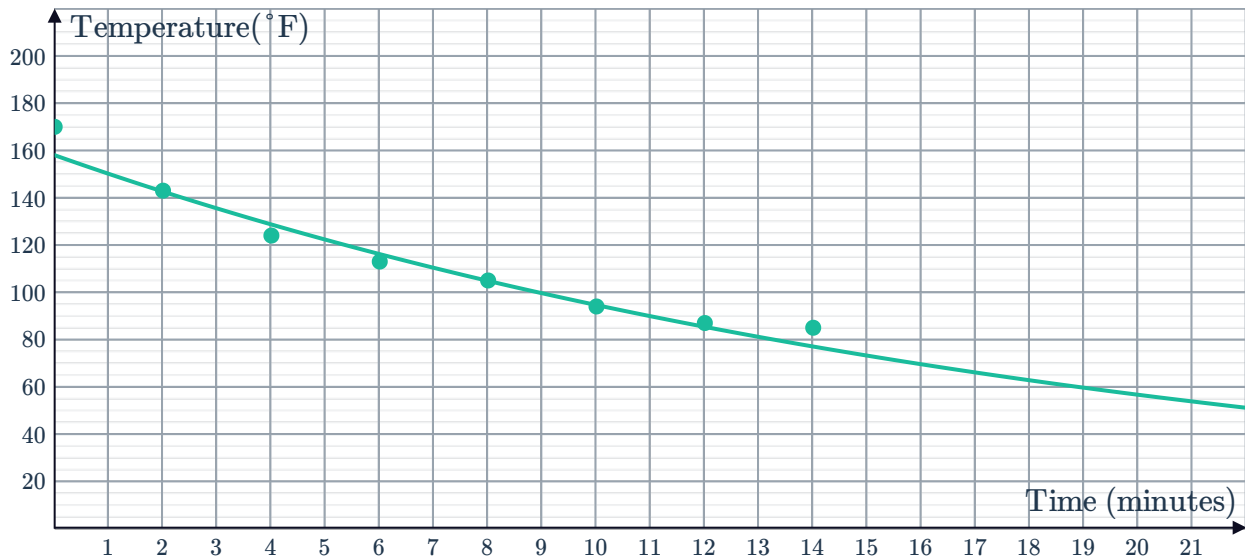
a We cannot approximate the new pan's quality rating by fitting a linear, exponential or quadratic function to the data.

b False

7



a



Yes, an exponential model is appropriate because when drawn on a graph it has an exponential shape which is decreasing less rapidly over time.

Also, if we calculate an exponential regression on this data set using the exponential function, we get  $R^2 = 0.9647$  which is very close to 1, so there is a strong correlation to the exponential model suggesting an exponential association.

- b  $y = 157.9684(0.9521^x)$ , where  $x$  is the number of minutes and  $y$  is temperature.
- c Most of the points corresponding to actual measurements lie on the graph of the function.
- d  $59.2^\circ \text{ F}$

8

a C

b

|     |   |        |
|-----|---|--------|
| $t$ | 0 | 10     |
| $y$ | 0 | 49 000 |

c 2 439 000

9

a  $M = 50\,000(1.032)^t$ b  $P = 49\,000(1.061)^t$ 

c \$9500.24

d Elizabeth's

10

a 8 million

b 7130 million

c 0.87%

11



- a 10
- b 20 %
- c 100 %
- d The population of bacteria in the dark environment is increasing at a faster rate than the population of bacteria in the sunlit environment.

12

- a  $A = P(1.053)^t$
- b  $B = P(1.01)^{4t}$
- c Bank A

13

- a The sale price of property.
- b 5.5%
- c -3.6%

14

- a  $P = 90 + 5t$
- b  $Q = 90 \times (1.04)^t$
- c the current company
- d opening his own business

15

- a  $P = (0.85)^t$
- b Maximilian
- c C
- d 13%
- e No

16

- a  $P = 2\,410\,000(1.022)^n$
- b  $F = 3\,374\,000(1.02)^n$
- c Yes
- d The food production is increasing, but at a slower rate than the population's rate of increase.
- e Over a decade from now.

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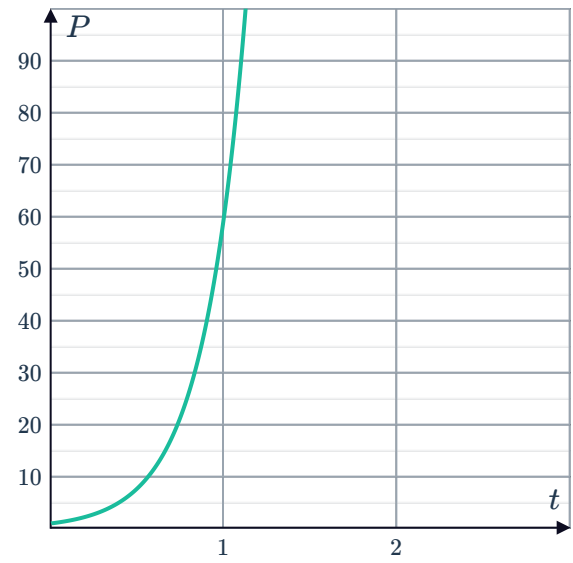
- a 179
- b 132
- c person B's
- d 0.7
- e if you have a high sugar diet, your blood sugar level peaks at greater amounts and decreases more rapidly than a low sugar diet

18

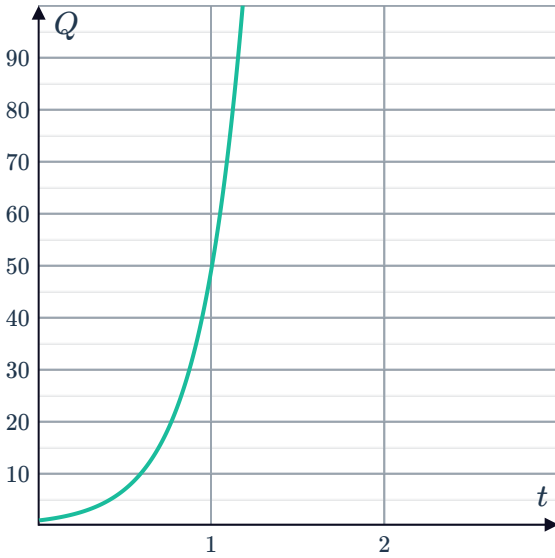
a Faster



b



c



d No